

CHRISTIAN BLEVENS

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ML and software engineer with production experience shipping deep learning systems end-to-end. UW Seattle BS in Electrical Engineering, Digital Signal & Image Processing concentration. Currently deploying a novel disentangled-VAE architecture at AudeAI and building a B2B SaaS in 3D scene reconstruction. Strong across PyTorch, FastAPI, React, Three.js, Rust/Bevy, and Unity.

EXPERIENCE

Machine Learning Engineer — AudeAI, Seattle, WA

Sept 2025 – Present

- Designed and trained a novel VAE architecture compressing text embeddings into a 10-dimensional disentangled latent space, with each dimension mapping to an interpretable communication trait.
- Achieved 0.65 cosine similarity to source embeddings at maximum compression; per-dimension KL targets verified near-full disentanglement.
- Deployed the model as an independently callable backend service and integrated it into the team's React-based AI evaluator.
- Lifted evaluator performance across accuracy, consistency, evidence retrieval, and advice quality; exposed latent dimensions to non-ML stakeholders for direct inspection.

Founder — Voluhaus, Seattle, WA

Jan 2026 – Present

- Built a B2B SaaS for the home-staging industry that reconstructs 3D room geometry from a single photo and places AI-generated furniture on real surfaces.
- Shipped the full stack solo: Three.js front-end, FastAPI backend, DuckDB, Modal for GPU inference, Cloudflare R2 storage, and a Hetzner production server behind nginx and Cloudflare.
- Integrated MoGe-2 for monocular 3D reconstruction; built surface detection and furniture placement on top of it.
- Ran customer discovery and direct cold outreach to staging companies; iterated pitch based on operator feedback.

Freelance Software Engineer — Self-Employed, Seattle, WA

June 2023 – Sept 2025

- Scoped, priced, and delivered freelance software engineering projects for clients sourced through online technical communities; ran each engagement end-to-end including client communication, design mockups, and delivery.
- Notable project: a web app that overlays user-drawn Christmas-light placements on a photo of a home, then uses depth maps, line detection, and AI image generation to render a realistic install preview.
- Produced visual mockups and documentation so non-technical clients could give meaningful feedback before implementation; advised on project viability and proposed alternative approaches when original scopes were unworkable.
- Concurrently built engineering portfolio of 30+ projects (christianblevens.me) spanning ML systems, real-time graphics, and full-stack web (the body of work that led directly to the AudeAI role).

Capstone Engineer — Echodyne (UW-sponsored), Seattle, WA

Jan 2023 – June 2023

- Contributed to a UW capstone sponsored by Echodyne, expanding their ML training image database via radar-driven automated labeling.
- Built the system converting radar object detections into pan/tilt/zoom commands for a tracking camera, including coordinate transforms between radar and camera frames.
- Validated coordinate math in Unity and Python simulations before deploying to hardware; owned subsystem network communications.
- Onboarded onto a multi-year existing codebase and contributed within team conventions.

EDUCATION

University of Washington — Seattle, WA

Graduated June 2023

B.S. in Electrical Engineering, concentration in Digital Signal & Image Processing

Relevant coursework: Machine Learning for Signal Processing, Random Signals for Communications, Digital Signals & Filtering, Modern Wireless Communications, Introduction to Digital Imaging Systems, Data Structures & Algorithms, Probability & Statistics for Engineers, Introduction to Computer Networks.

SELECTED PROJECTS (FULL PORTFOLIO AT CHRISTIANBLEVENS.ME)

Stock Sentiment Predictor. Pre-norm transformer fusing 1D-conv-encoded price tokens with attention-weighted subject-channel sentiment from news and Reddit. Per-horizon delta and confidence heads at 1d/1w/1m. FastAPI service, GPU AMP, Docker deploy.

Live Audio Censoring. Real-time browser-mic profanity censor with sample-accurate playback ducking. Streaming Whisper word-level timestamps rebased onto frontend recording time; Web Audio gain automation; auto-calibrated end-to-end delay.

AI Video Editor. Multi-track editor with AI transcription, pause/filler/profanity detection, and burn-in word-level subtitles. Batched FastAPI backend running ~12.5x real-time; GPU-accelerated FFmpeg export at multiple quality presets.

Arcane Physiology + Roughcast (Bevy + wgpu). Multi-component Lattice Boltzmann fluid simulation running entirely on the GPU via wgpu compute shaders; five immiscible fluids with Shan-Chen inter-component repulsion drive the magic system of a full game. Merged with a destructible multi-area voxel world (GPU Surface Nets meshing over a Lipschitz-tree-pruned SDF primitive graph for $O(\log n)$ per-voxel evaluation) and a 2M-blade GPU-instanced grass system over arbitrary tagged surfaces. Unified Bevy ECS bridges editor state, fluid compute passes, voxel meshing, and gameplay.

SKILLS

Languages: Python, Rust, C#, C++, JavaScript/TypeScript, Java

ML / AI: PyTorch, VAEs, diffusion & flow matching, transformers, embeddings, RAG, SHAP, sentence-transformers, Qdrant

Web / Backend: React, Three.js, FastAPI, DuckDB, Modal, nginx, Cloudflare, Docker, GPU inference deployment

Graphics / Systems: Bevy, wgpu compute shaders, Unity DOTS/ECS, Rapier3D, voxel systems, real-time rendering

DSP / EE: Digital signal & image processing, Fourier analysis, radar-camera coordinate fusion, FPGA fundamentals

Tools: Git, agentic coding workflows, Docker Compose, Linux server admin